
Computer Networks Lab

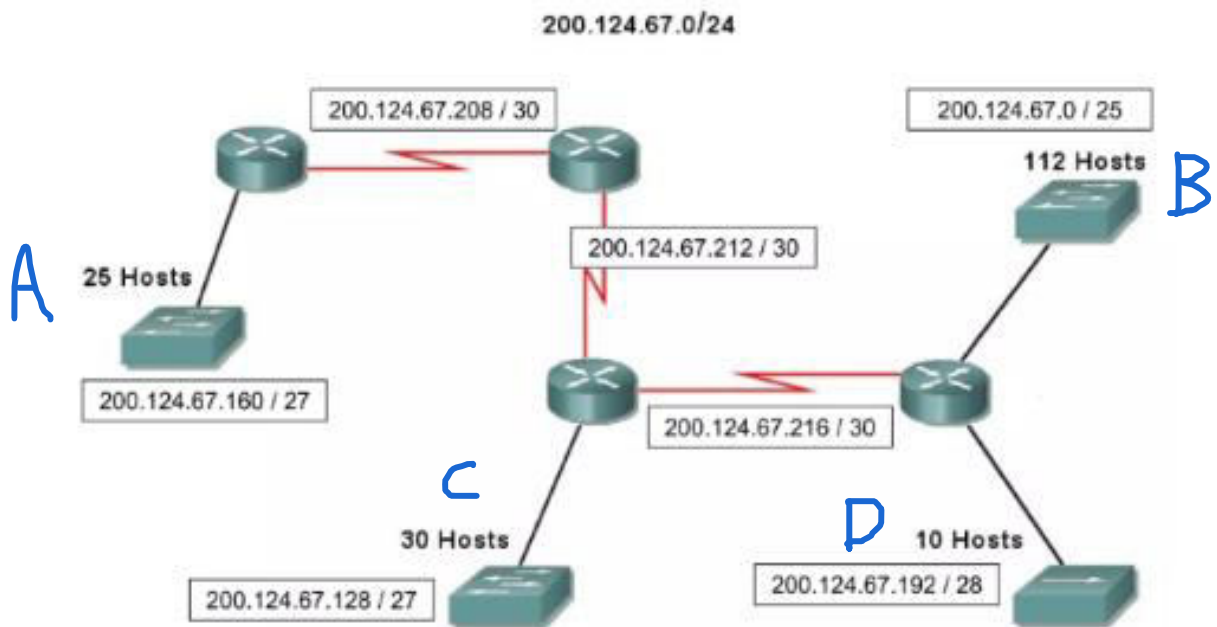
Fall 2025
Week 11

Variable Length Subnet Mask (VLSM)

Lab Task

Task 1:

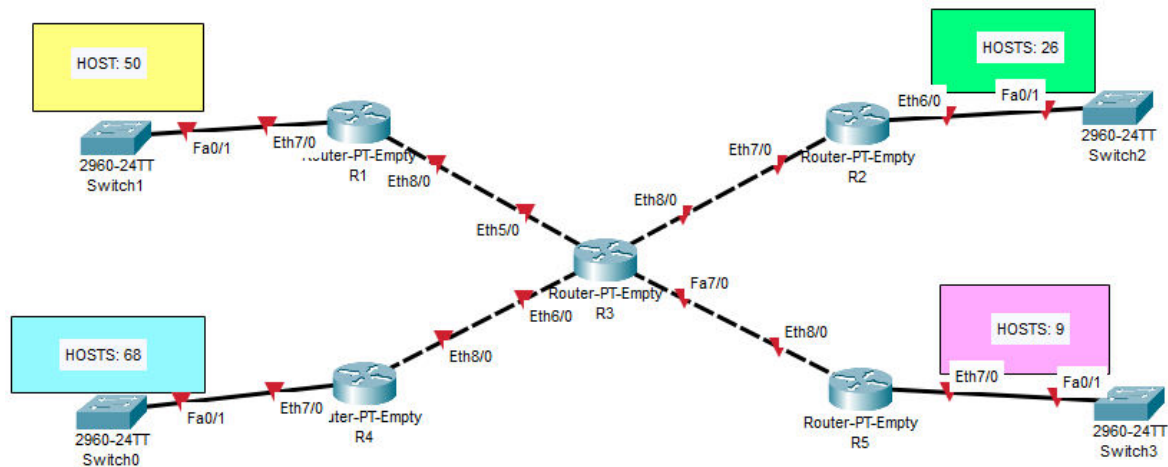
Design the vlsn network below:



You have to show all your working by creating either VLSM Tree or table showing the ranges of subnets and subnet masks along with Network address and Broadcast address of each. After this create a topology as shown above and assign all ip addresses.

Task 2:

Let's say you have a network with the address 200.10.20.0/24. You need to create subnets for the given topology. You have to show all your working by creating either VLSM Tree or table showing the ranges of subnets and subnet masks along with Network address and Broadcast address of each. Then create a topology as shown below and assign all ip addresses.



Submission Instructions:

Two files are required for submission:

- **Cisco Packet Tracer File (.pkt):**
 - File name format: Section_RollNumber.pkt
 - Must contain complete working topology with all configurations saved
- **Documentation Report (Word or PDF):**
 - File name format: Section_RollNumber.docx or .pdf
 - Must include: VLSM Subnetting working, Network topology diagram, router configurations on CLI, routing tables (show ip route screenshots), connectivity testing (minimum 4 ping/pdu tests).

ALL screenshots in your report must display your Roll Number using the Note tool in Cisco Packet Tracer. Screenshots without roll numbers will not be accepted.